

SPOORTHI BASU

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Distributed Systems & Real-Time Data Infrastructure Engineer

PROFESSIONAL EXPERIENCE

Software Engineer, [Genesys Cloud Services](#), March 2021 – Present

- Designed and built Flink pipelines transforming company-wide Kafka streams into schema-driven Iceberg/S3 datasets at hundreds of millions of events/day, powering customer-facing analytics and self-service downloads.
- Architected and built the company's first end-to-end data validation framework, validating correctness from event ingestion through Flink/Iceberg to customer queries; catches incidents before customers, adopted by 3 teams.
- Scaled fault-tolerant Java/Kafka microservices to 10M+/day via partitioning/rebalancing, cutting latency by 25%.
- Engineered customer-facing REST APIs at 5K+ RPS with Redis cache sharding, halving DB load.
- Built org-wide integration testing 0→1 (LocalStack/AWS mocking), automating test creation to <1min for 5+ teams.
- Delivered 99.99% uptime through multi-region deployment and circuit breakers (Resilience4j) for fault isolation.
- Drove delivery of 10+ customer-facing features with Product and QA, growing daily active users by 25%.
- Mentored junior engineer on distributed systems, halving onboarding time while maintaining 15-min P0/P1 SLA.

Software Engineer, [Coding Minds](#), July 2020 – February 2021

- Shipped a full-stack academic platform (React/Java/Node.js) on Heroku serving 500+ daily active users.
- Developed RESTful APIs (Java/Spring) over MySQL, hardened to 95% test coverage before release.
- Cut page load time 30% by adding React memoization and lazy loading across the app.

PUBLICATIONS & OPEN SOURCE

InfoQ Article: [The Schema Proliferation Problem in Kafka and Flink Pipelines: How to Solve It](#)

- Authored a deep-dive on InfoQ that ranked #1 Top Article in the weekly Software Development Round-Up, read by a community of 550K+ global senior developers monthly, peer-reviewed before publication.
- Diagnosed schema proliferation in Kafka/Flink/Avro pipelines as the cost of one-to-one event-to-schema mapping, where overlapping event types fan out into separate schemas and Iceberg tables, requiring cross-table joins.
- Introduced a discriminator-based consolidation pattern using enum routing fields and nullable attribute blocks, collapsing 12+ event schemas into a single table and eliminating the cross-table UNION queries it had forced.
- Designed a layered adapter that decouples transformation logic from Flink integration, with a reproducible open-source reference implementation on GitHub that readers can run end to end on a local Kafka/Flink stack.

Apache Flink CDC – Open Source Contributions | [PR #4360](#) | [PR #4418](#)

- PR #4360 (merged, shipping in Flink CDC 3.7): fixed duplicate records in the Iceberg sink when a schema change splits writes mid-checkpoint; commits each batch as its own snapshot so equality-deletes supersede stale rows.
- PR #4418 (open): fixes a JobManager OOM on large MySQL CDC tables, where the source coordinator retained snapshot-split metadata after entering the binlog phase; releases it once the split is safely checkpoint-covered.
- The merged fix was reviewed and approved by an Apache Flink CDC committer, with end-to-end and unit tests covering schema-change, failover, and retry edge cases.

TECHNICAL SKILLS

- **Languages & Frameworks:** Java, Python, JavaScript, SQL, C++, Spring Boot (MVC, JPA), React
- **Data & Streaming:** Apache Kafka, Apache Flink, Apache Iceberg, Avro, CDC, Real-Time Data Pipelines
- **Cloud & DevOps:** AWS (Lambda, EC2, S3, DynamoDB, SQS), Redis, Docker, Terraform, Jenkins, CI/CD
- **Testing & Monitoring:** JUnit, Mockito, Testcontainers, LocalStack, Postman, New Relic, Sumo Logic
- **Core Concepts:** Distributed Systems, System Design, Concurrency, Object-Oriented Design (OOD), Test-Driven Development (TDD), Data Structures & Algorithms, Fault Tolerance
- **AI-Assisted Development:** Cursor, Claude Code, Amazon Q, LLM-assisted code review and test generation

EDUCATION

Master's in Computer Science, California State Polytechnic University Pomona (May 2020) - GPA: 3.66

Bachelor's in Computer Science, Dr. Ambedkar Institute of Technology, India (June 2018) - GPA: 4.0